CHIEN-HSUN CHANG 張健勳

Backend / Platform Engineer | AI Systems / MLOps
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Taiwan – Open to roles across Taiwan and remote work
Available for 2027 pre-employment programs and R&D Substitute Service



Education

National Chung Cheng University Expected Aug. 2027
M.S. in Computer Science and Information Engineering

National Taichung University of Science and Technology Jun. 2025
B.Eng. in Intelligent Production Engineering
Graduated 1st in department; Academic Excellence Award (5 times); Outstanding Graduate Award

Summary

Backend / Platform Engineer with experience in AI Systems and MLOps. Built backend services, ML inference workflows, Kubernetes deployments, CI/CD pipelines, and Linux infrastructure. Contributed to CNCF KServe and worked on digital twin, computer vision, and federated learning projects.

Experience

National Chung Cheng University **Aug. 2025 – Present**
Research Assistant / Project Lead

- Lead a 4-member international team building a flood disaster digital twin platform with FastAPI, PostgreSQL/PostGIS, Redis, and Docker.
- Build APIs, spatial data pipelines, and ML inference workflows connecting simulation, GIS, decision support, and visualization.
- Set up CI/CD, automated deployment, and validation workflows with GitHub Actions.

System Administrator

- Maintain Linux servers, GPU workstations, Docker environments, networking, storage, SSH access, and shared research infrastructure.

PAIA Technology Co., Ltd. **Jul. 2024 – Jun. 2025**
Backend Software Engineer Intern

- Developed backend services with Python, Django Ninja, Pydantic, PostgreSQL, and MongoDB.
- Built REST APIs, validation logic, database models, logging, and error handling.
- Wrote unit and integration tests and participated in code review and Agile/Scrum development.

Additional Experience: Programming Instructor, CodingAPE Mar. 2023 – Jun. 2025

Selected Projects

KServe (CNCF) – Open Source Contributor **Sep. 2025 – Present**
Python · Go · Kubernetes · CRD · Helm

- Fixed a Python logging issue in KServe; the patch was merged upstream.
- Submitted `runtimeClassName` support for ServingRuntimePodSpec and WorkerSpec, including tests and CRD/OpenAPI updates.

TAG-Twin Flood Disaster Digital Twin Platform **Aug. 2025 – Present**
FastAPI · PostgreSQL/PostGIS · Redis · Docker · WebSocket · GIS

- Built the core backend and data platform integrating GIS data, flood simulation, risk data, evacuation routing, and Unreal Engine visualization.
- Built data and ML inference pipelines, APIs, and WebSocket interfaces connecting simulation, decision, and visualization modules.

Cross-Nation Federated AQI Platform **Aug. – Sep. 2026**
PyTorch · Flower · Kubernetes · Kubeflow · KServe · Grafana

- Led system planning and cross-team integration for the CCU × UBM federated AQI platform, defining interfaces and handoff contracts across dataset preparation, Flower training, deployment, inference, and monitoring.
- Integrated the federated global model with the UBM AI Deployment Platform, including AQI inference, Grafana monitoring, end-to-end testing, and final deployment.

AI Environmental Enforcement System **Sep. 2025 – Present**
PyTorch · YOLO11 · ByteTrack · Qwen3-VL · TensorRT · Docker

- Designed and validated a two-stage CCTV event pipeline: YOLO11x detection, ByteTrack + OSNet Re-ID, pose rules, Qwen3-VL review, and plate OCR.
- Separated GPU inference from rerunnable post-processing and deployed the system with the industry partner for field use.

Skills

Languages: Python, Go, Rust, C/C++, SQL
Backend & Data: FastAPI, Django Ninja, Pydantic, PostgreSQL/PostGIS, MongoDB, Redis, REST API, WebSocket
AI / ML: PyTorch, scikit-learn, Computer Vision, Federated Learning, ML Inference
Platform / MLOps: Linux, Docker, Kubernetes, KServe, Kubeflow, Helm, GitHub Actions, Git/GitLab, Grafana, Prometheus
Digital IC: Verilog, RTL design, functional simulation, APR, DRC/LVS; TSMC N16 AD FP design flow coursework